MIS

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Roadmap

Background

- ☒ Technology
- ☒ Infrastructure
- ☒ IS and its core components
- ☒ Technology estimates in finance, accounting, marketing, and management
- ☒ Role of Data
- ☒ IS versus IT

Hardware, Software, and Networks

- ☒ Hardware components
  - ☒ Processing
  - ☒ Storage
  - ☒ Peripherals, IoT, and wearables
  - ☒ Mobile
- ☒ Software
  - ☒ Systems software versus Application software (Apps)
  - ☒ Licensing: Perpectual, Subscription, and Open Source
  - ☒ Cloud Computing: IaaS, PaaS, SaaS, AIaaS, QaaS
- ☒ Networks
  - ☒ LAN, WAN, VPN, WiFi, 5G (Cellular)
  - ☒ Topologies: Star, Mesh, Bus
  - ☒ Internet: TCP/IP

Version Control Systems

- ☒ GitHub
  - ☐ using Git Bash
- ☒ Google Docs
- ☒ Dropbox

Excel

- ☒ Data: extraction, formatting, and cleaning
- ☒ Formula and Statistic presentation
  - ☒ Statistical formulas:
    - ☒ General: AVERAGE(), SUM(), MEDIAN(), STDEV(), PERCENTILE(), MIN(), MAX(), SMALL(), LARGE()
    - ☒ Frequency: COUNT(), COUNTA(), COUNTBLANK()
    - ☒ Group-wise: COUNTIF(), COUNTIFS()
  - ☒ Utility formulas:
    - ☒ General: LAG()
    - ☒ Date: YEAR(), MONTH(), DAY(), WEEKDAY()
    - ☒ Conditionals: IF(), IFS(), AND(), OR(), ISBLANK(), IFNA()
    - ☒ String: LEFT(), RIGHT(), MID(), LEN(), UPPER(), LOWER(), PROPER(), &, TEXTJOIN(), FIND(), SUBSTITUTE()
- ☒ Aggregation and Search: VLOOKUP(), HLOOKUP()
- ☒ Search: MATCH()
- ☒ DCF Exercise

Case Studies

- ☐ Knight Capital
- ☐ Waddell Reed E-Mini Futures
- ☐ Amazon and Toys-R-Us
- ☐ Spotify and Apple Music
- ☐ Target Pregnancy Prediction
- ☐ Capital One Cloud Services
- ☐ Equifax Data Breach
- ☐ Hugging Face Breach - OpenAI

Assigned Case Studies

- ☐ CrowdStrike Outage
- ☐ McDonald's AI Drive Through
- ☐ Twitter/X Infrastructure Under Musk
- ☐ The MGM Resorts Ransomware Attack
- ☐ The Air Canada Chatbot
- ☐ Amazon's AI Recruiting Engine

SQL

- ☐ Data: creation and extraction
- ☐ Querying:
  - ☐ WHERE with Conditionals:
    - ☐ General: AND, OR, NOT, >, <, >=, <=, <>
    - ☐ Missing: NULL
    - ☐ List and range: IN (), BETWEEN *number1* AND *number2*
    - ☐ String:
      - ☐ LIKE [_ (underscore), % (percent)]

- ☐ GLOB [? (question mark), * (star)]
- ☐ SUBSTR(*variable_name*, *starting_position*, *number_of_characters*)

- ☐ SORT:
  - ☐ ORDER BY [ASC, DESC]
- ☐ Optional:
  - ☐ HAVING

☐ Multi-level Querying

☐ Computation and transformation

- ☐ Statistical formulas:
  - ☐ General: AVG(), SUM(), MEDIAN(), PERCENTILE(), MIN(), MAX()
  - ☐ Frequency: COUNT()
  - ☐ Group-wise: GROUP BY
- ☐ Utility:
  - ☐ General: LIMIT
  - ☐ Conditional cases: CASE WHEN-THEN-ELSE END AS *variable_name*
    - ☐ Single CASE statement
    - ☐ Multiple CASE statements
    - ☐ Nested CASE statements
  - ☐ Creating Tables:
    - ☐ Permanent using CREATE TABLE *table_name* AS
    - ☐ Temporary using WITH *table_name* AS ()
  - ☐ Dropping Tables:
    - ☐ DROP TABLE *table_name*
    - ☐ DROP TABLE IF EXISTS *table_name*
  - ☐ MERGE: LEFT JOIN, RIGHT JOIN, INNER JOIN, FULL JOIN